ROBOT INTELLIGENCE LAB

XAI506 Robot Intelligence Final Project

Interactive Pick and Place System

Interactive Pick and Place

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Our goal is to develop an **interactive pick-and-place system** for semi-humanoid robots.


Pick Target
target
left: can (can_4)
right: can (can_11)
pick
off
on
FFW SG2 Interactive Pick and Place
Mode choose_to_pick
Pick off
Selected left: can (can_4)
sim time [0.60]sec
wall time [3.83]sec
Time diff [3.23]sec

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◉ Project Takeaway

- Through this project, students should understand that pick-and-place is not just a single motion command, but a **full perception-to-control pipeline**.
- **Perception becomes a 3D target.**
  - RGB-D observations and segmentation masks can be combined to estimate object-centered 3D pick and place targets.
- **Grasping needs orientation, not only position.**
  - Matching only the TCP position is often not enough. Multiple site targets, such as TCP, top, side, and front points, help preserve the gripper orientation during IK.
- **IK solver settings affect motion quality.**
  - Parameters such as step size, iteration count, nullspace control, and joint-limit handling can change convergence speed, stability, smoothness, and final posture.
- **Pick-and-place is a phase sequence.**
  - A reliable behavior requires structured phases such as approach, pregrasp, grasp, lift, place approach, release, and retreat, rather than directly moving from pick to place.

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◉ Rule of Thumb

- “**Use observations that are directly accessible in real-world robotic systems.**”
- Available Observations
  - RGB-D camera observations
  - Multi-view camera inputs
  - Robot joint states
  - …
- Restricted Observations
  - Ground-truth object poses
  - Simulator-only privileged information
  - …

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◉ Methodology

- **Any approach that achieves interactive pick-and-place is valid.**
- Examples
  - Vision-Language-Action (VLA) policies
  - Reinforcement Learning (RL) policies
  - Foundation model-based approaches
  - Diffusion policies
  - Modular robotics pipelines
  - End-to-end visuomotor policies
  - …

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◉ Evaluation

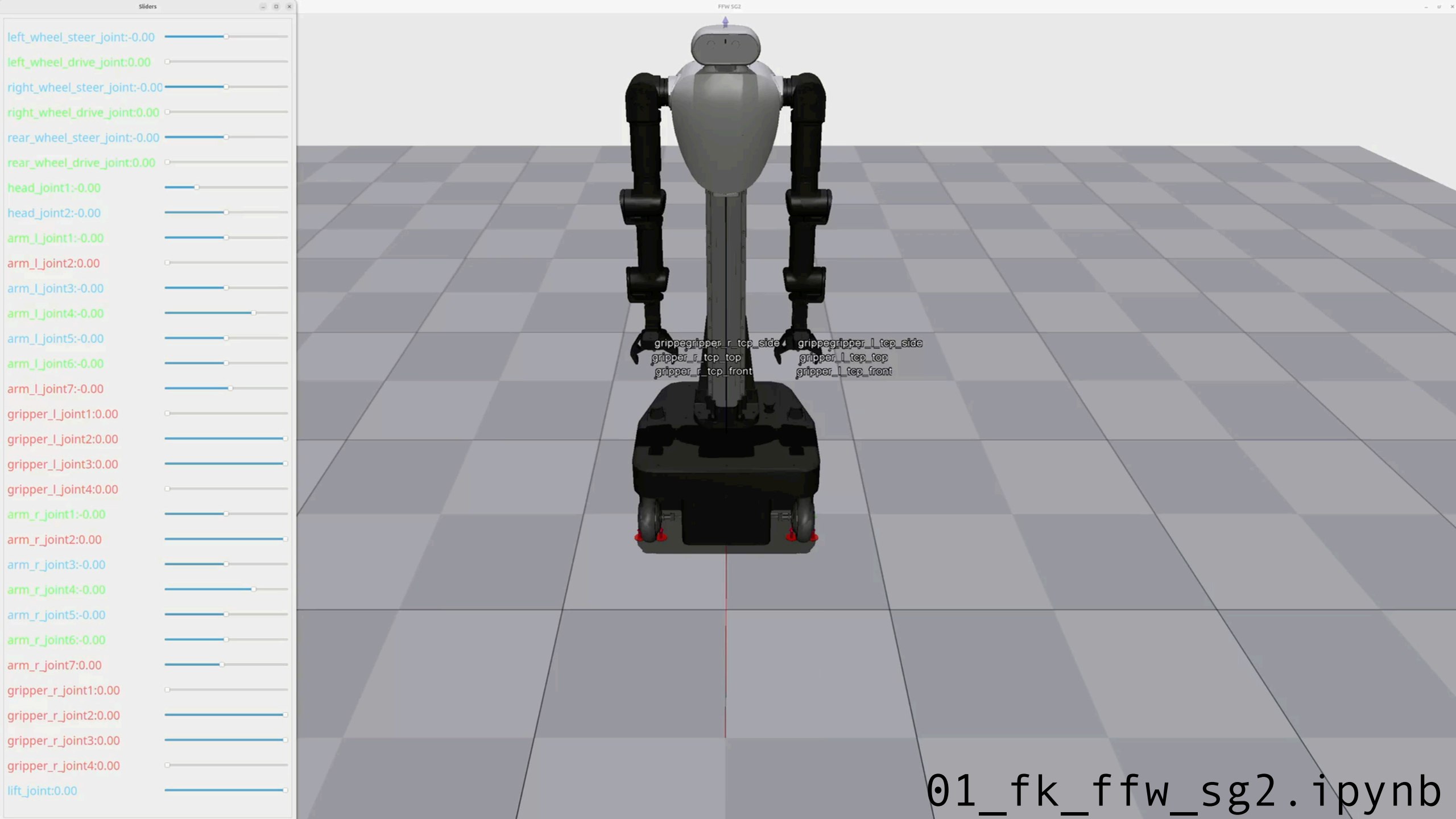
- The evaluation tests whether the system can perform pick-and-place from observation, rather than relying on fixed object poses or a memorized motion.
- (50%) We will run randomized tabletop trials and measure success rate in two settings:
  - **Seen objects**:10 trials with objects used during development.
  - **Unseen objects**:10 trials with new objects not used during development.
  - Object and place-target positions will be randomized within a reachable tabletop region.
  - Success is measured by whether the selected object is picked, moved, and placed at the target region.
- (50%) **Final presentation (26.06.17, 5 mins)**
  - Discuss the approaches explored for interactive pick-and-place and the challenges encountered during development.
  - Share lessons learned from perception, grasping, and motion execution experiments.

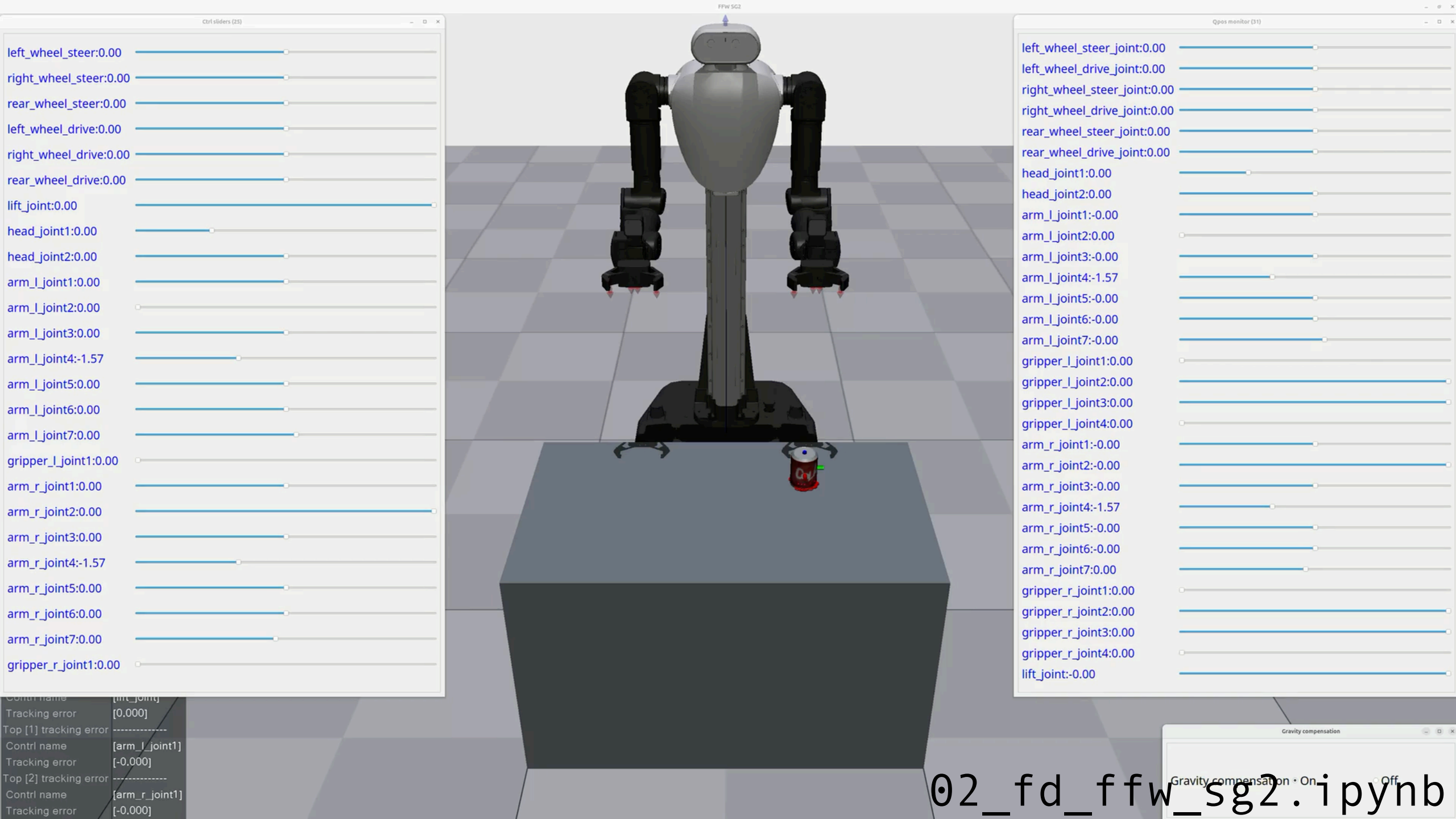
Interactive Pick and Place

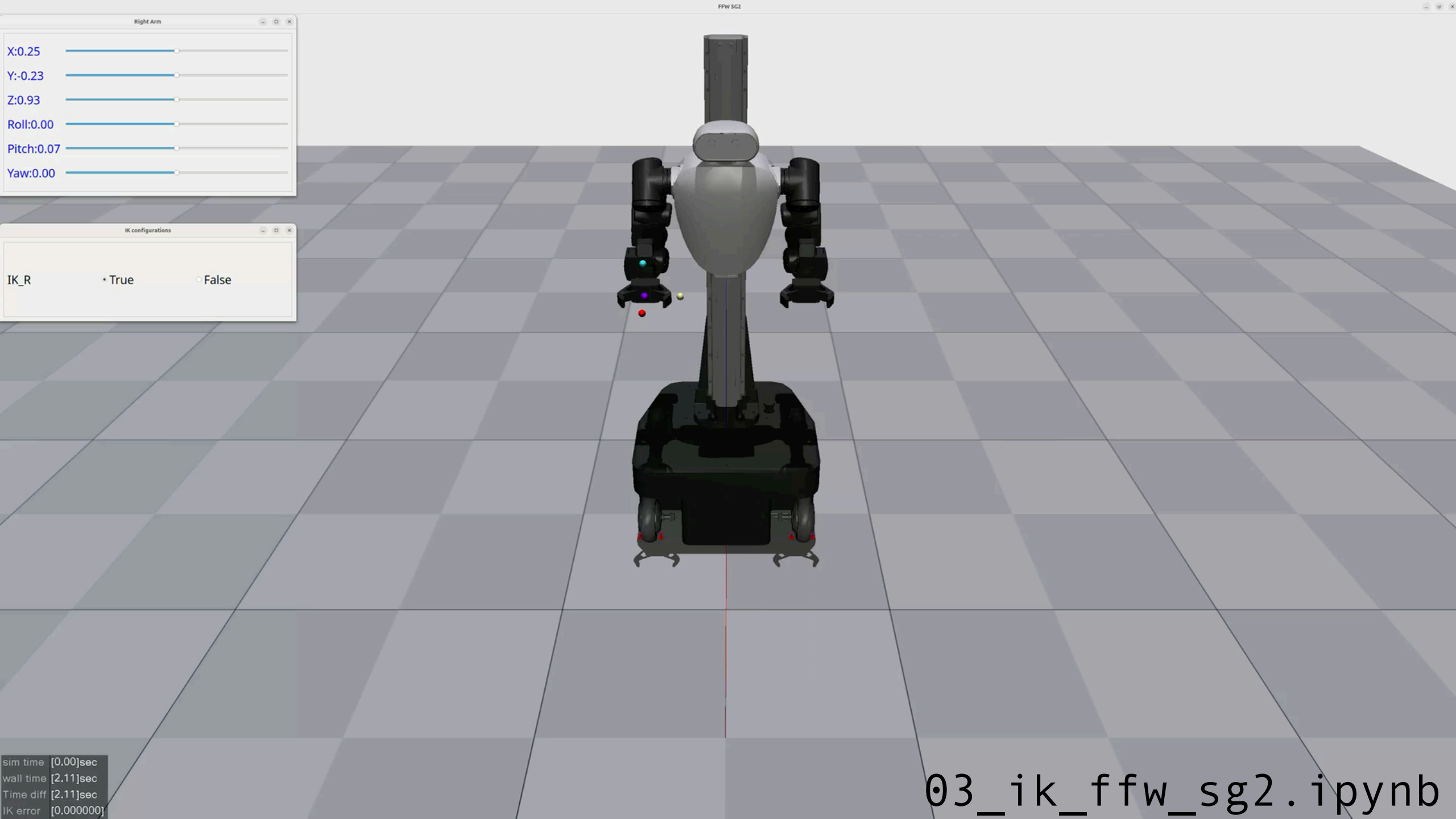
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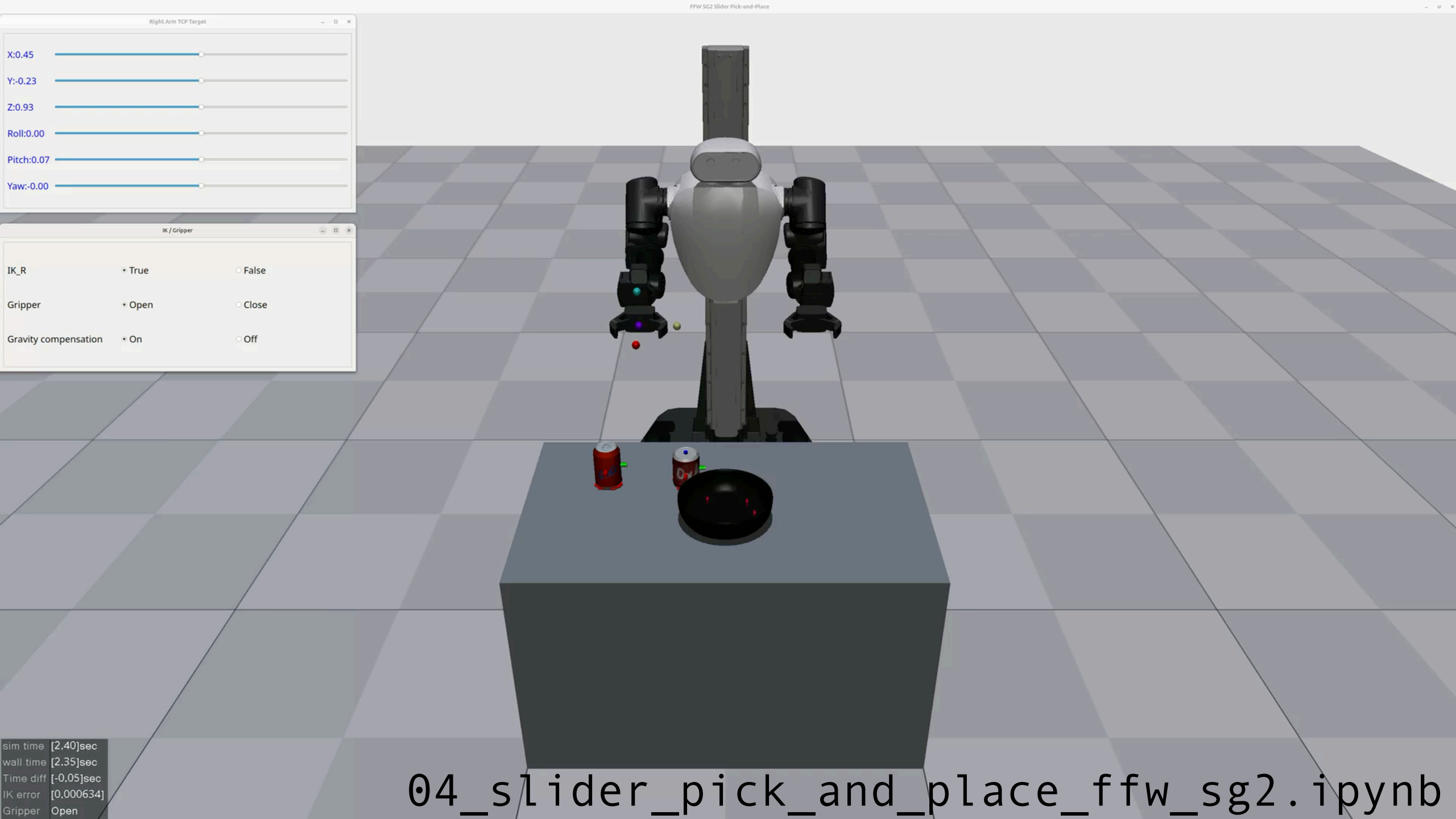
◉ What are Given?

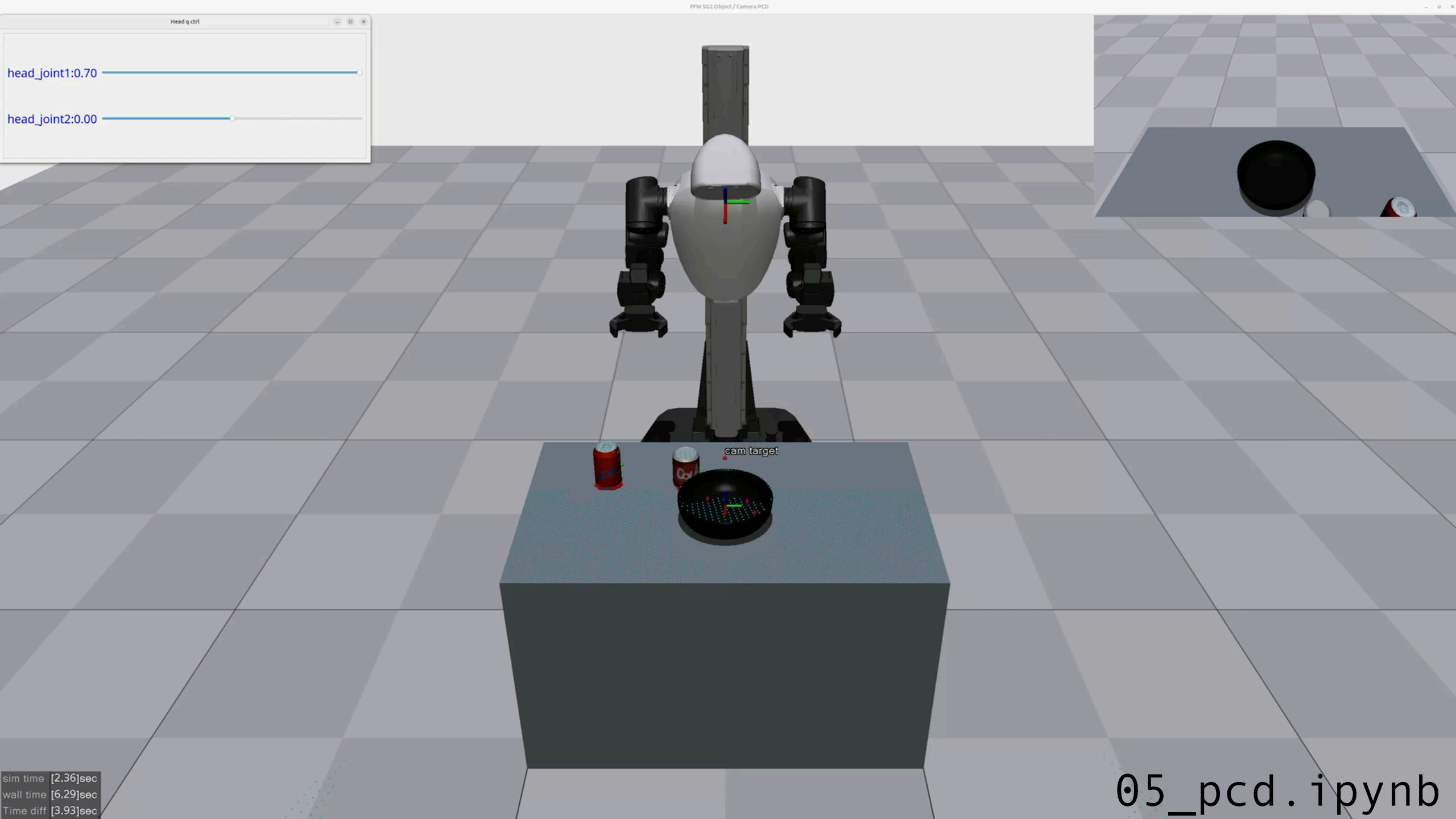
- 01_fk_ffw_sg2.ipynb
- 02_fd_ffw_sg2.ipynb
- 03_ik_ffw_sg2.ipynb
- 04_slider_pick_and_place_ffw_sg2.ipynb
- 05_pcd.ipynb
- 06_hf_owlv2_sam2.ipynb


Sliders
left_wheel_steer_joint:-0.00
left_wheel_drive_joint:0.00
right_wheel_steer_joint:-0.00
right_wheel_drive_joint:0.00
rear_wheel_steer_joint:-0.00
rear_wheel_drive_joint:0.00
head_joint1:-0.00
head_joint2:-0.00
arm_l_joint1:-0.00
arm_l_joint2:0.00
arm_l_joint3:-0.00
arm_l_joint4:-0.00
arm_l_joint5:-0.00
arm_l_joint6:-0.00
arm_l_joint7:-0.00
gripper_l_joint1:0.00
gripper_l_joint2:0.00
gripper_l_joint3:0.00
gripper_l_joint4:0.00
arm_r_joint1:-0.00
arm_r_joint2:0.00
arm_r_joint3:-0.00
arm_r_joint4:-0.00
arm_r_joint5:-0.00
arm_r_joint6:-0.00
arm_r_joint7:0.00
gripper_r_joint1:0.00
gripper_r_joint2:0.00
gripper_r_joint3:0.00
gripper_r_joint4:0.00
lift_joint:0.00
FFW SG2
grippegripper_r_tcp_side
gripper_r_tcp_top
gripper_r_tcp_front
grippegripper_l_tcp_side
gripper_l_tcp_top
gripper_l_tcp_front
01_fk_ffw_sg2.ipynb


FFW SG2
Ctrl sliders (25)
left_wheel_steer:0.00
right_wheel_steer:0.00
rear_wheel_steer:0.00
left_wheel_drive:0.00
right_wheel_drive:0.00
rear_wheel_drive:0.00
lift_joint:0.00
head_joint1:0.00
head_joint2:0.00
arm_l_joint1:0.00
arm_l_joint2:0.00
arm_l_joint3:0.00
arm_l_joint4:-1.57
arm_l_joint5:0.00
arm_l_joint6:0.00
arm_l_joint7:0.00
gripper_l_joint1:0.00
arm_r_joint1:0.00
arm_r_joint2:0.00
arm_r_joint3:0.00
arm_r_joint4:-1.57
arm_r_joint5:0.00
arm_r_joint6:0.00
arm_r_joint7:0.00
gripper_r_joint1:0.00
Qpos monitor (31)
left_wheel_steer_joint:0.00
left_wheel_drive_joint:0.00
right_wheel_steer_joint:0.00
right_wheel_drive_joint:0.00
rear_wheel_steer_joint:0.00
rear_wheel_drive_joint:0.00
head_joint1:0.00
head_joint2:0.00
arm_l_joint1:-0.00
arm_l_joint2:0.00
arm_l_joint3:-0.00
arm_l_joint4:-1.57
arm_l_joint5:-0.00
arm_l_joint6:-0.00
arm_l_joint7:-0.00
gripper_l_joint1:0.00
gripper_l_joint2:0.00
gripper_l_joint3:0.00
gripper_l_joint4:0.00
arm_r_joint1:-0.00
arm_r_joint2:-0.00
arm_r_joint3:-0.00
arm_r_joint4:-1.57
arm_r_joint5:-0.00
arm_r_joint6:-0.00
arm_r_joint7:0.00
gripper_r_joint1:0.00
gripper_r_joint2:0.00
gripper_r_joint3:0.00
gripper_r_joint4:0.00
lift_joint:-0.00
Tracking error [0.000]
Top [1] tracking error
Contrl name [arm_l_joint1]
Tracking error [-0.000]
Top [2] tracking error
Contrl name [arm_r_joint1]
Tracking error [-0.000]
Gravity compensation
Gravity compensation On Off
02_fd_ffw_sg2.ipynb


FFW SG2
Right Arm
X:0.25
Y:-0.23
Z:0.93
Roll:0.00
Pitch:0.07
Yaw:0.00
IK configurations
IK_R
True
False
sim time [0.00]sec
wall time [2.11]sec
Time diff [2.11]sec
IK error [0.000000]
03_ik_ffw_sg2.ipynb

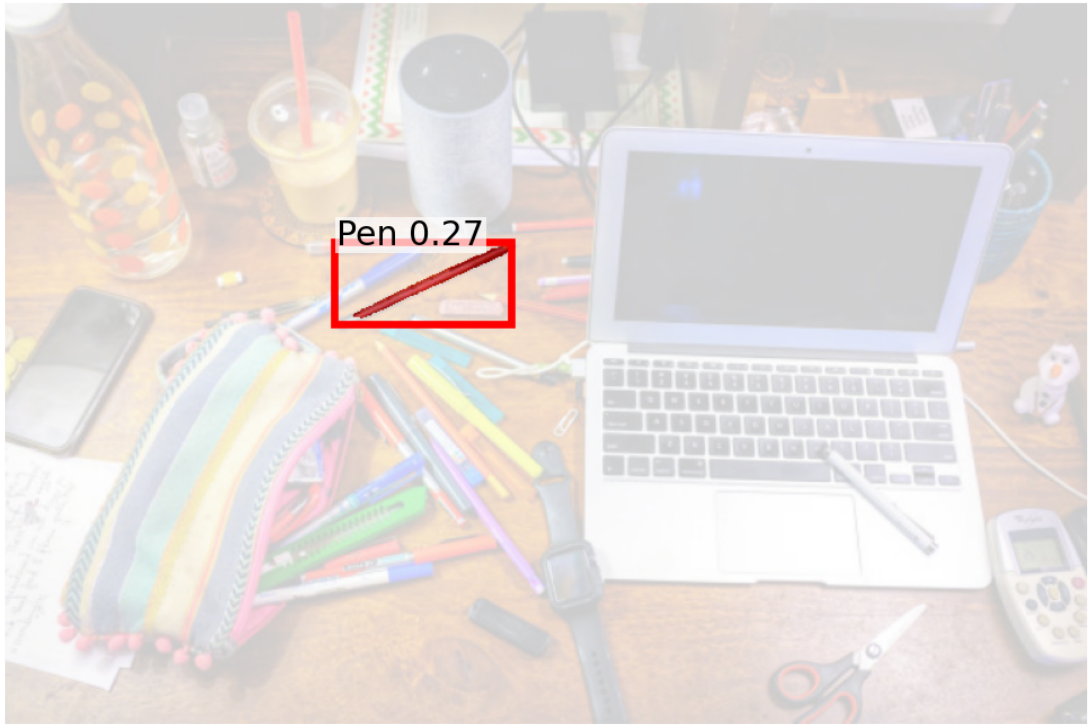

FFW SG2 Slider Pick-and-Place
Right Arm TCP Target
X:0.45
Y:-0.23
Z:0.93
Roll:0.00
Pitch:0.07
Yaw:-0.00
IK / Gripper
IK_R
True
False
Gripper
Open
Close
Gravity compensation
On
Off
sim time [2.40]sec
wall time [2.35]sec
Time diff [-0.05]sec
IK error [0.000634]
Gripper Open
04_slider_pick_and_place_ffw_sg2.ipynb


FFW SG2 Object / Camera PCD
Head q ctrl
head_joint1:0.70
head_joint2:0.00
cam target
sim time [2.36]sec
wall time [6.29]sec
Time diff [3.93]sec
05_pcd.ipynb

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06_hf_owlv2_sam2.ipynb

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How?

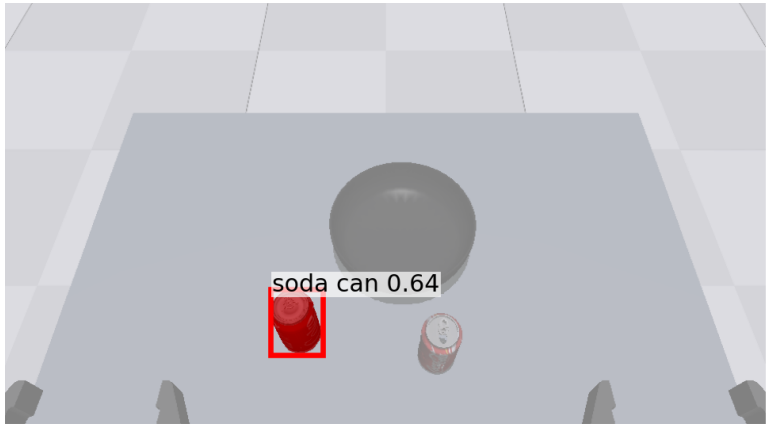

Pick Target
target
left: can (can_4)
right: can (can_11)
pick
off
on
FFW SG2 Interactive Pick and Place
Mode choose_to_pick
Pick off
Selected left: can (can_4)
sim time [0.60]sec
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Interactive Pick and Place

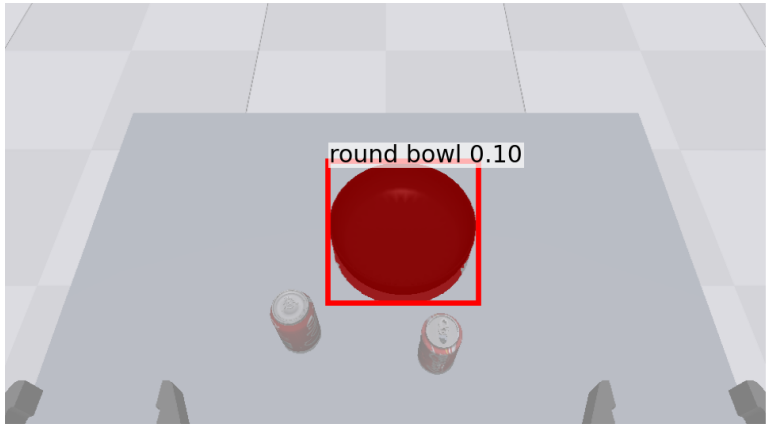
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◉ Step 1.

- **Localize pick-and-place targets using foundation models**
  - Prompt-to-bounding-box (OWL-ViT / OWLv2)
  - Prompt-to-point prediction (Molmo2)
  - Point-to-segmentation (SAM2)
  - Point, bounding box, and trajectory generation (Gemini ER 1.6)
  - …







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◉ Step 2.

- Estimate pick-and-place target poses from RGB-D point clouds

◉ Step 3.

- Execute the motion using inverse kinematics

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◉ Code Structure

- **pick_and_place_utils.py** contains stateless helper functions for scene setup, RGB-D perception, point-cloud processing, segmentation, grasp target generation, and simple gripper commands.
  - build_point_cloud_observation
    - Compute a 3D target point from the segmented object point cloud.
  - get_grasp_offsets
    - Define guide points around the gripper TCP to help preserve orientation during IK.
  - make_pick_place_phase_targets
    - Build the phase-wise targets for approach, grasp, lift, place, and retreat.
  - get_gripper_ctrl
    - Convert open/close commands into left/right gripper actuator controls.
- **pick_and_place_core.py** contains stateful APIs for observation collection, IK solving, and the interactive pick-and-place controller.
  - DEFAULT_SOLVER_CONFIG
    - Set the IK solver hyperparameters that affect convergence speed, stability, smoothness, and posture behavior during motion.
  - update_pose_phase
    - Use the current phase target, solve IK, set gripper commands, and decide when to move to the next phase.
  - make_execute_step_info
    - Package IK and gripper controls into the command dictionary used by env.step(...).

Thank You for Your Attention

